Adding all these equations and cancelling the terms on the Left Hand side, we get,

$$(n+1)^3 - 1^3 = 3(1^2+2^2+3^2+\cdots+n^2) + 3(1+2+3+\cdots+n) + n$$

$$n^3 + 3n^2 + 3n = 3(1^2+2^2+3^2+\cdots+n^2) + \frac{3n(n+1)}{2} + n$$

$$3(1^2+2^2+3^2+\cdots+n^2) = n^3 + 3n^2 + 2n - \frac{3n(n+1)}{2} = \frac{2n^3+6n^2+4n-3n^2-3n}{2}$$

$$3(1^2+2^2+3^2+\ldots+n^2) = \frac{2n^3+3n^2+n}{2} = \frac{n(2n^2+3n+1)}{2} = \frac{n(n+1)(2n+1)}{2}$$

$$1^2+2^2+3^2+\cdots+n^2 = \frac{n(n+1)(2n+1)}{6}$$

2.11.4 Sum of cubes of first *n* natural numbers

To find $1^3+2^3+3^3+\cdots+n^3$, let us consider the identity

$(x+1)^4 - x^4 = 4x^3 + 6x^2 + 4x + 1$

Where x = 1, 2, 3,.....n – 1, n

$$x = 1,\ 2^4 - 1^4 = 4(1)^3 + 6(1)^2 + 4(1) + 1$$

$$x = 2,\ 3^4 - 2^4 = 4(2)^3 + 6(2)^2 + 4(2) + 1$$

$$x = 3,\ 4^4 - 3^4 = 4(3)^3 + 6(3)^2 + 4(3) + 1$$

$$\vdots \qquad \vdots \qquad \vdots$$

$$x = n-1,\ n^4 - (n-1)^4 = 4(n-1)^3 + 6(n-1)^2 + 4(n-1) + 1$$

$$x = n,\ (n+1)^4 - n^4 = 4n^3 + 6n^2 + 4n + 1$$

Adding all these equations and cancelling the terms on the Left Hand side, we get,

$$(n+1)^4 - 1^4 = 4(1^3+2^3+3^3+\cdots+n^3) + 6(1^2+2^2+3^2+\cdots+n^2) + 4(1+2+3+\cdots+n) + n$$

$$n^4 + 4n^3 + 6n^2 + 4n = 4(1^3+2^3+3^3+\cdots+n^3) + 6\times\frac{n(n+1)(2n+1)}{6} + 4\times\frac{n(n+1)}{2} + n$$

$$4(1^3+2^3+3^3+\cdots+n^3) = n^4 + 4n^3 + 6n^2 + 4n - 2n^3 - n^2 - 2n^2 - n - 2n^2 - 2n - n$$

$$4(1^3+2^3+3^3+\cdots+n^3) = n^4 + 2n^3 + n^2 = n^2(n^2+2n+1) = n^2(n+1)^2$$

$$1^3+2^3+3^3+\cdots+n^3 = \left(\frac{n(n+1)}{2}\right)^2$$



Ideal Friendship

Consider the numbers 220 and 284.

Sum of the divisors of 220 (excluding 220) = 1+2+4+5+10+11+20+22+44+55+110=284

Sum of the divisors of 284 (excluding 284) =1+2+4+71+142=220.

Thus, sum of divisors of one number excluding itself is the other. Such pair of numbers is called Amicable Numbers or Friendly Numbers.

220 and 284 are least pair of Amicable Numbers. They were discovered by Pythagoras. We now know more than 12 million amicable pair of Numbers.

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